



NYU

TANDON SCHOOL
OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 1 · Introduction to the Bootcamp & Python Fundamentals

Instructor: **Angelin Ann Jacob**

Section 1 · Tue Sept 29

Section 2 · Thu Oct 1



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Part 1 • Introduction to Bootcamp

Get to know your program and timeline for the next 9 weeks!



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Meet the team! - TCH Data Science Bootcamp



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Nine weeks, two sections

Week	Topic	Section 1 · Tue	Section 2 · Thu
1	Introduction to Bootcamp & Python Fundamentals	Sept 29	Oct 1
2	NumPy & Pandas	Oct 6	Oct 8
3	Exploratory Data Analysis & Data Visualization	Oct 13	Oct 15
4	SQL Fundamentals	Oct 20	Oct 22
5	Mid-Program Project Presentations	Oct 27	Oct 29
6	Machine Learning Fundamentals, Perceptron	Nov 3	Nov 5
7	Probability, Naive Bayes & Linear Regression	Nov 10	Nov 12
8	Linear & Logistic Regression, Bias-Variance Trade-off & Gradient Descent	Nov 11	Nov 19
–	No sessions · Thanksgiving break	–	–
9	Final Project Presentations	Dec 1	Dec 3

Stay with the section you registered for: Tue 11:00 AM–1:00 PM ET or Thu 1:00–3:00 PM ET

How each session runs

01

Lecture

Concepts and examples through slides and Jupyter notebooks

02

Hands-on

Apply it right away: real datasets, Python, analysis and visualization

03

Q&A

Ask anything, anytime. The more you ask, the more you learn

04

Take-home

Weekly problems on the topic, due one week later

05

Breaks

Several short breaks through each two-hour session

The goal every week: theory plus practice you can apply to **real-world data science projects**.

Useful Links

Zoom

All sessions are virtual. Log in with your NYU credentials so attendance counts.

Brightspace

Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub

Slides and Colab notebooks for every session.

Email

datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

1:1 coaching

Optional 30-minute technical interview prep. [Book it through Handshake.](#)

Bootcamp social

Mon, Oct 5, 3–5 PM · 2 MetroTech Center, Room 1054. Drinks, snacks, teammates.

Useful Links

Brightspace

Assignments, submissions and the discussion board, monitored Monday to Thursday.

Data Science Bootcamp · Fall 2026

The screenshot shows the NYU Brightspace interface. At the top is the NYU logo and a navigation bar with icons for grid, mail, chat, notifications, and a user profile for Khushi Agarwal. Below the navigation bar are links for Calendar, Announcements, NYU Bookstore, Help, and Course Feedback. A large purple banner reads "Welcome to NYU Brightspace, Khushi".

The main content area is divided into two columns. The left column, titled "My Courses", shows a list of courses with filters for 2023 Fall, 2022 Spring, 2022 Fall, 2021 Fall, and "Project Sites" (which is selected). The courses listed are:

- TCH Bootcamps: Data Science - Section 1 (3-877-50768-50770-TANDON PROJECT • _Project Sites)
- TCH Bootcamps: LeetCode Section 1 (4-877-50768-50770-TANDON PROJECT • _Project Sites)
- TCH Bootcamps: LeetCode - Section 2 (PRJ-TANDON-SU26-071726-06 • _Project Sites)
- TCH Bootcamps: Data Science - Section 2 (PRJ-TANDON-SU26-071726-07 • _Project Sites)

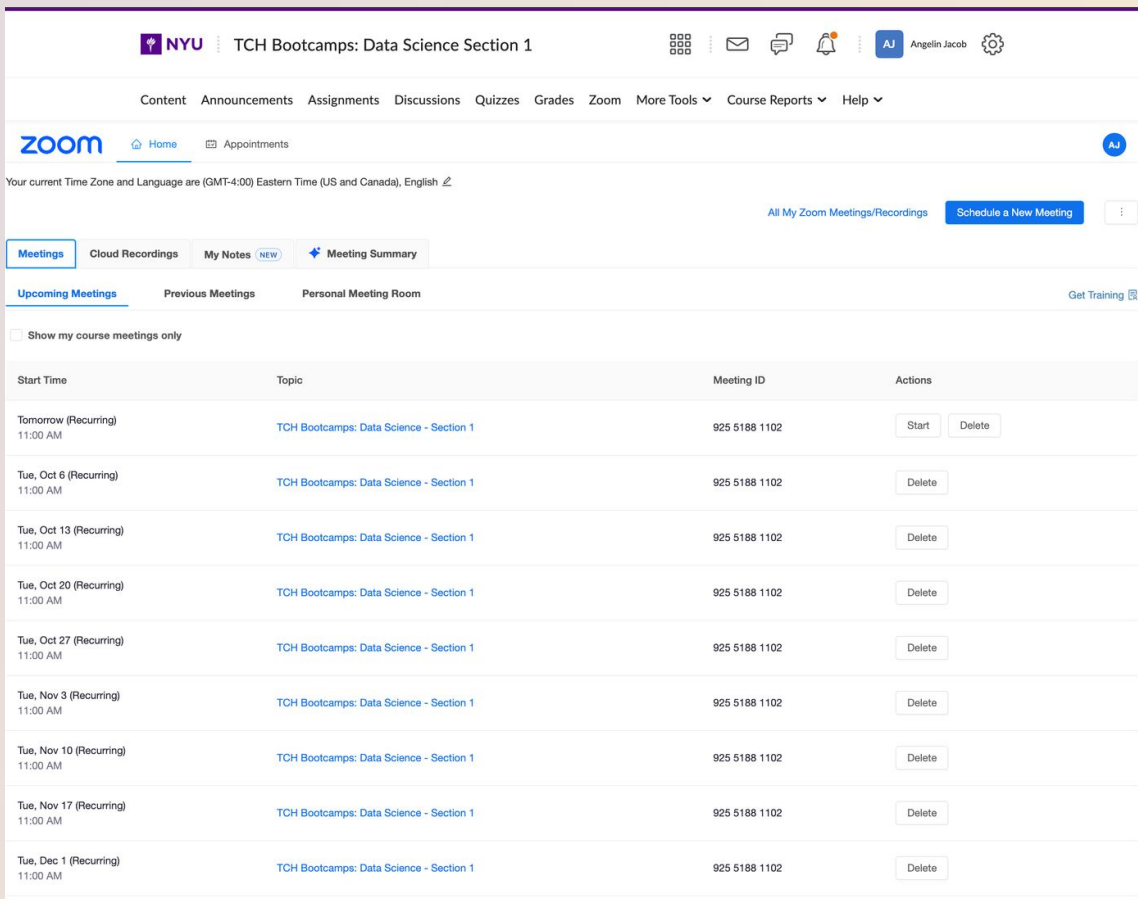
The right column contains a "Work To Do" section for the period "September 24 - October 7", showing a task for "PHP HW" due on Oct 4. Below this is an "Announcements" section titled "NYU Brightspace Overview" with a video thumbnail labeled "Navigate NYU Brightspace" and "NYU BRIGHTSPACE OVERVIEW".

Useful Links

Zoom

All sessions are virtual.

Log in with your NYU credentials so attendance counts. Please attend 7 out of 9 sessions to confirm attendance.



The screenshot shows the Zoom web interface for a user named Angelin Jacob. The page title is 'TCH Bootcamps: Data Science Section 1'. The navigation bar includes links for Content, Announcements, Assignments, Discussions, Quizzes, Grades, Zoom, More Tools, Course Reports, and Help. The Zoom logo and 'Home' link are visible. Below the navigation bar, there's a section for 'Your current Time Zone and Language are (GMT-4:00) Eastern Time (US and Canada), English'. A button 'Schedule a New Meeting' is present. The main content area shows a list of upcoming meetings under the 'Upcoming Meetings' tab. The list includes a table with columns for Start Time, Topic, Meeting ID, and Actions.

Start Time	Topic	Meeting ID	Actions
Tomorrow (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Start</button> <button>Delete</button>
Tue, Oct 6 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Oct 13 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Oct 20 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Oct 27 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Nov 3 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Nov 10 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Nov 17 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>
Tue, Dec 1 (Recurring) 11:00 AM	TCH Bootcamps: Data Science - Section 1	925 5188 1102	<button>Delete</button>

Useful Links

Brightspace Assignment Submission Page

- All assignments submission will happen through Brightspace.
- All files will be submitted via a **.ipynb Jupyter Notebook**.
- Please follow the guidelines given within the Assignment for naming conventions of files:
`netID_week_1.ipynb`
- This weeks assignment is due on **Oct 6th, 2026**

NYU

TCH Bootcamps: Data Science Section 1

AJ

Angelin Jacob

Content

Announcements

Assignments

Discussions

Quizzes

Grades

Zoom

More Tools

Course Reports

Help

+ New Unit

Visible

Add Existing

Create New

Search titles, descriptions

Getting Started

Week 1 - Introduction to Bootcamp and Python Fundamentals

Week 1 Information

Week 1 Take Home Assignment
Due: Oct 6, 2026 11:59 PM

Week 1 Feedback Form
Due: Oct 6, 2026 11:59 PM

Week 1 - Introduction to Bootcamp and Python Fundamentals

Resources

Week 2 - Numpys and Panda

Week 3 - Exploratory Data Analysis and Data Visualization

Week 4 - SQL Fundamentals

Week 5 - Mid Program Project Presentations

Take-Home Submission Guidelines

File type:

For every assignment, we will be using a notebook, which can consist of a mix of Markdown and Code cells. Please submit a single notebook in the .ipynb format, any other format will not be accepted by the auto-grader.

File name

Name your notebook with your **NetID** and the **week number** in this format: `netid_week_number.ipynb`

Example naming convention:

Week	Example (NetID ab1234)
Week 1	ab1234_week_1.ipynb

- Use **lowercase**, with an **underscore** between NetID, "week" and week number (no spaces).
- Submit **one notebook per week** on Brightspace, by 11:59 PM one week post each lecture. Week 1's assignment is due on **Oct 6th, 2026**.

Notebook Cell names

Each exercise will be labelled with its respective markdown cells in the format `## Exercise <number>: <task_name>`

Ensure to put your code in the cell(s) directly below it.

What you'll be given within the boilerplate notebook.

```
## Exercise 1: Vowel counter
```

Answer format layout example:

<code>## Exercise 1: Vowel counter</code> <code>[your code]</code>	↔ Markdown cell ↔ Code cell(s)
<code>## Exercise 2: Upper case</code> <code>[your code]</code>	↔ Markdown cell ↔ Code cell(s)

...

Refer to the Resources section of Week 1 to read more on handling ipynb notebooks if you are unfamiliar with it.

Before you submit

- File is named `netid_weeknumber.ipynb`
- Every answer/code cells sits under its respective `## Exercise <number>: <task_name>` markdown cell.
- **Kernel › Restart & Run All** runs without errors, and outputs are visible
- Skipped an exercise? Keep the heading and write **Not attempted** below it

less

Earning the digital credential

A verifiable Accredible badge you can share on LinkedIn, once you meet all five:

Attend 7+ live sessions

Attendance is logged by time on Zoom. Please **ONLY** use your [NYU Zoom](#) SSO login/credentials to access Zoom. Arriving late or leaving early needs prior approval by email, or it counts as absent.

Submit every take-home

On Brightspace, due one week after it's assigned, by 11:59 PM on your section's day.

Send weekly feedback

Fill out the teaching section feedback form after each session.

Form a project group

3–4 members. Submit names and your chosen prompt by **Sunday, Oct 18, 11:59 PM.**

Present twice

Mid-program on Oct 27/29 and final on Dec 1/3, with every group member present.

The project

Groups of 3–4 pick one real-world prompt and present their work to industry professionals. Prizes go to the winning group.

**Data Visualization with Air
Quality Data**

**Sentiment Analysis of
Social Media Content**

**Movie Recommendation
System**

**Finance & Economics
Time-Series Analysis**

KEY DATES

Mon, Oct 5

Bootcamp social: meet people, form groups

Sun, Oct 18

Group names and prompt due, 11:59 PM

Oct 27 / 29

Mid-program presentation and feedback

Dec 1 / 3

Final presentation

Three roles in data

Today touches all three: we ingest and clean raw form data, summarize it, and form hypotheses.

Data Analyst

FOCUS

Organizing data and identifying patterns

TYPICAL WORK

Reports, dashboards, summary statistics

CORE SKILLS

Python, SQL, visualization

Data Engineer

FOCUS

Building the infrastructure that moves and stores data

TYPICAL WORK

Collection tools, storage, processing pipelines

CORE SKILLS

Python, SQL, ETL, distributed systems

Data Scientist

FOCUS

Modeling complex data to support decisions

TYPICAL WORK

Statistical analysis, machine learning, experimentation

CORE SKILLS

Python, statistics, ML, visualization



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Part 2 • Introduction to Python Fundamentals

Today's dataset is you! 127 bootcamp applications.



Who's in the room

127

applications to this
cohort

54%

call themselves
beginners (68 of 127)

80%

are master's students
(101 of 127)

72%

already list Python as
a language they know
(92)

Most common major: **Computer Science** (48) · Most common time budget: **5–7.5 hours a week** (81)

Source: bootcamp application form, 127 anonymized responses

Today's agenda

Each part introduces one group of language features and uses it on the data

Part	Python concept	Question it answers
0	Working in a notebook	How do we run code?
1	Variables and primitive types	What kind of data did the form collect?
2	Operators and expressions	How do we tell a late-night submission from an early-morning one?
3	Lists, tuples, dictionaries, sets	What does a single application look like?
4	Conditionals and loops	When did people apply? Who skipped questions? What do we study?
5	String, list and dictionary methods	When did everyone submit? Which languages do we already know?
6	Functions and error handling	Can we build a small analytics toolkit? What excites applicants most?
7	Scope	Where do variables live, and when do they disappear?

Before next week

Take-home

The exercises at the end of today's notebook. Submit on Brightspace within one week, by 11:59 PM.

Feedback form

Fill it out after today's session. It counts toward your credential.

Week 2: NumPy & Pandas

Everything we did today with loops and dictionaries, pandas does in a line or two.

See you at the social: **Mon, Oct 5, 3–5 PM** · 2 MetroTech Center, Room 1054